Structural Equation Models

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Introduction

Structural equation modelling (SEM) is the unified framework that integrates measurement (how observed items reflect underlying latent constructs) and structure (how latent constructs relate to each other) into a single estimation problem. It generalises confirmatory factor analysis (measurement only), path analysis (regressions among observed variables only), and multivariate regression. Modern psychometrics, organisational research, and increasingly clinical-outcomes work depend on SEM as the primary analytical tool because it isolates relationships among true constructs from those distorted by measurement error.

Prerequisites

A working understanding of confirmatory factor analysis, multiple regression, latent-variable thinking, and basic familiarity with `lavaan` syntax.

Theory

SEM estimates measurement and structural parameters jointly by maximum likelihood (or robust ML, weighted least squares for ordinal indicators). Components in `lavaan` syntax:

- **Measurement model** (`=~`): latent factor defined by its observed indicators.
- **Structural regressions** (`~`): latent-on-latent or latent-on-observed regressions.
- **Variances / covariances** (`~~`): residual correlations or factor covariances.
- **Defined parameters** (`:=`): derived quantities such as indirect effects.

Standard fit indices (CFI, TLI, RMSEA, SRMR) parallel those of CFA. Indirect effects and confidence intervals are computed at the level of MCMC or bootstrap draws.

Assumptions

Correct specification of both the measurement and structural components, multivariate Normal continuous indicators (or robust estimators for non-Normal data), and sufficient sample size — at least 200 observations as a rough rule, more for complex models.

R Implementation

```r
library(lavaan); library(semPlot)

model <- '
  # Measurement
  ind60 =~ x1 + x2 + x3
  dem60 =~ y1 + y2 + y3 + y4
  dem65 =~ y5 + y6 + y7 + y8

  # Structural
  dem60 ~ ind60
  dem65 ~ ind60 + dem60
```

```r
  # Residual correlations
  y1 ~~ y5
  y2 ~~ y4 + y6
  y3 ~~ y7
  y4 ~~ y8
  y6 ~~ y8
'
fit <- sem(model, data = PoliticalDemocracy)
summary(fit, standardized = TRUE, fit.measures = TRUE)

semPaths(fit, "std", layout = "tree2")
```

Output & Results

`summary(fit, standardized = TRUE)` reports unstandardised and standardised coefficients, factor loadings, residual covariances, and the full set of fit indices. `semPaths()` renders a path diagram with arrow widths proportional to standardised coefficients.

Interpretation

A reporting sentence: "The structural equation model fit the Political Democracy data well (CFI 0.99, RMSEA 0.04 [90 % CI 0.02 to 0.06], SRMR 0.05); industrialisation in 1960 predicted both 1960 and 1965 democracy ($\beta = 0.45$ and 0.39 respectively), and 1960 democracy strongly predicted 1965 democracy ($\beta = 0.84$, $p < 0.001$)." Always report standardised coefficients alongside fit indices.

Practical Tips

- Check the measurement model in isolation (run a CFA) before adding structural paths; weak factors produce uninterpretable structural relationships.
- Use `modificationIndices()` cautiously; data-driven modifications blur the confirmatory status of the analysis and inflate Type I error.
- For mediation analysis, define indirect effects with `:=` and request bootstrap CIs (`sem(..., se = "bootstrap")`); the indirect effect is a product of coefficients with non-Normal sampling distribution.
- For ordinal indicators, switch to weighted least squares (`estimator = "WLSMV"`); ordinary ML on Likert items violates Normality assumptions.
- Very complex models need very large samples; if $n < 5$ per estimated parameter, results are unstable and a simpler path model may be more robust.
- For cross-cultural or multi-group comparisons, fit measurement-invariance hierarchies (configural, metric, scalar) before comparing latent means across groups.

R Packages Used

`lavaan` for `sem()`, `cfa()`, `growth()`, and `lavInspect()`; `semPlot` for path diagrams; `semTools` for measurement-invariance tests, reliability coefficients, and modification heuristics; `blavaan` for Bayesian SEM that handles small samples and complex priors.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**

- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE